

Imaging of the Spine

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Along with a comprehensive history and physical examination, imaging is essential for the evaluation of the patient who presents with suspected or known spine pathology. It facilitates preoperative assessment of a patient, generation of a differential diagnosis, selection of the most likely diagnosis, postoperative evaluation, and monitoring of disease progression. Unlike radiologists, most orthopaedic surgeons have little, if any, formal training in the systematic evaluation and interpretation of imaging studies. On the other hand, the orthopaedic surgeon has access to information that the radiologist does not, such as a patient's history and physical examination, which can affect diagnosis and surgical decision-making. Experience may help the clinician decide which imaging modality is best for a patient, evaluate imaging studies effectively, determine the most likely diagnosis, and use the imaging studies in conjunction with clinical findings to guide surgical decision-making. We suggest that a team approach in conjunction with a radiologist will help the surgeon systematically evaluate each imaging study, correlate studies with the clinical information more accurately, reduce the incidence of unnecessary imaging, and increase the likelihood of making the correct and clinically relevant diagnosis and treatment decisions.

The purpose of this chapter is to review the imaging modalities currently used for evaluation of the spine in terms of evolving technique, image quality, speed, cost, availability, and safety, with a specific focus on the evaluation of degenerative diseases of the spine.

IMAGING MODALITIES

Radiographs

Radiographs involve the use of ionizing radiation and a radiation detector. Radiography can be discussed in detail based on the medium used as a radiation detector. In conventional radiography (CR), a film is used as a radiation detector. The film undergoes chemical processing after radiation exposure before the final image appears on the film and can be viewed against a bright light to illuminate the target anatomy. In digital radiography (DR), a digital detector is exposed to radiation transmitted through the patient. The imaging plate and image processor are used to convert radiation energy to light energy and eventually to a digital image, which can be stored and viewed on a computer. DR has been further subclassified into different forms based on

the type of detector used. Computed radiography, the initial type of DR developed in the early 1990s, uses a photo-stimulable phosphor plate, whereas direct radiography, developed in the late 1990s to early 2000s, uses semiconductor-based sensors, such as selenium, or selenium and cesium iodide.¹ The images created via DR can be directly transferred to picture archiving and communication system (PACS). PACS has a wide range of functions from image acquisition, digital display windows, storage of image, and generation of hard copies.

Until as late as the 1990s, a conventional film cassette was the primary method of acquiring radiographic images of the bones, but digital image acquisition has replaced the cassette method at almost all institutions. Advantages of DR include improved image quality, speed, accessibility, less radiation exposure, availability of postimaging processing and quality optimization, lower cost, and ease of image storage and retrieval.

Radiography (conventional or digital) is a widely available and cost-effective modality for the initial evaluation of the spine; it is valuable for trauma assessment, determination of coronal and sagittal deformity, identification of spondylosis and spondylolisthesis (and their progression), and detection of osteolytic and osteoblastic lesions suggestive of malignancy. Initial evaluation often begins with anteroposterior (AP) and lateral views of the area of interest (Fig. 124.1). The need for additional studies, such as oblique, flexion, or extension views, is determined by the clinical situation; for example, stress radiographs may be obtained to evaluate instability, which may be seen in patients with ligamentous injuries, or tension radiographs would be recommended in the evaluation of scoliosis. Radiographs can be used to determine the level of abnormality preoperatively and intraoperatively, and to provide a rapid evaluation of hardware placement and deformity correction intraoperatively and postoperatively.

Although radiography provides a relatively effective assessment of the osseous structures and their alignment, it is limited in its ability to visualize soft tissues, the spinal cord, the occipitocervical and cervicothoracic junctions, and bone marrow involvement. Furthermore, reductions in bone mass are evident on radiographs only after a 30% to 50% decrease in bone mineral density.^{2,3} Therefore radiography is not the most sensitive technique for the evaluation of bone pathologies, such as osteoporosis, which are due to decrease bone density.

Abstract

The purpose of this chapter is to review the imaging modalities currently used for evaluation of the spine in terms of evolving technique, image quality, speed, cost, availability, and safety, with a specific focus on the evaluation of degenerative diseases of the spine. The chapter focuses on multiple modalities for imaging of the spine, which includes conventional radiography, EOS2D/3D images, computed tomography, magnetic resonance imaging, myelography, nuclear scintigraphy, and functional metabolic imaging. Key principles for the analysis of degenerative disc disease are also outlined.

Keywords

spine
imaging
x-ray
EOS2D/3D
CT
MRI
myelography
SPECT/CT

EOS 2D/3D X-Ray Imaging System

EOS is a biplane x-ray imaging system manufactured by EOS imaging. It uses slot-scanning technology in which the radiation source and detector move in different planes during image acquisition. The slot-scanning technology can image the full length of the body up to 175 cm, and thus remove the need to manually join images. The quality of images is similar in nature to CR.⁴ It can take posteroanterior (PA) and lateral images simultaneously and construct a three-dimensional (3D) mode, which can allow analysis of vertebral rotation, flexion, translation, or rotation of scoliotic curve. The current software is effective for imaging patients of all ages, but 3D constructions cannot

be made for children younger than 6 years of age.⁵ Indications for the use of EOS are in diseases that change under load, such as scoliosis, lordosis, or kyphosis. It is increasingly being used for spine, pelvis, lower limb, and gait analysis imaging, and is considered a good technique for assessing global spinal sagittal balance and its relationships with the pelvis and lower limbs (Fig. 124.2).⁶

EOS produces high-quality images with less radiation than CR. Studies comparing EOS with conventional x-rays showed that image quality was comparable or slightly better with EOS. EOS images were superior or equivalent to computed radiography in image quality and structure visibility in 97.2% and 94.3% of images, respectively.⁷ Additionally, radiation dose was significantly

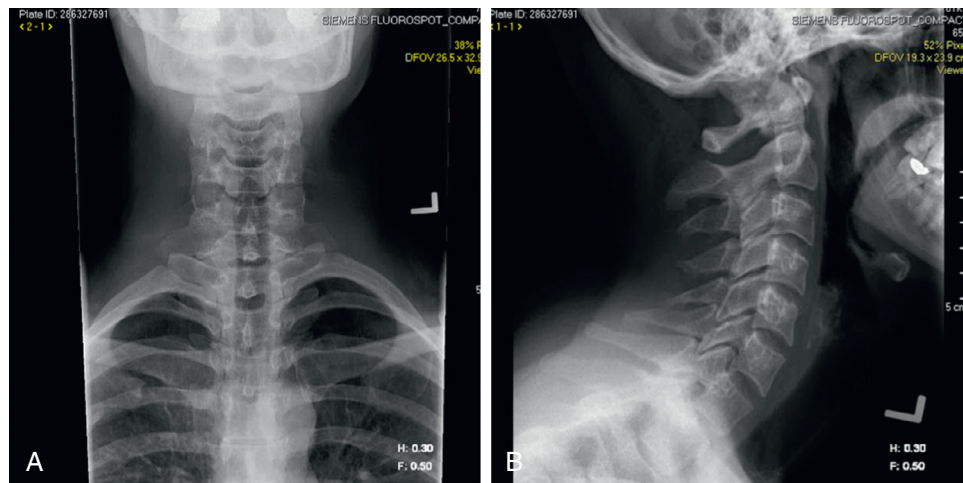


Fig. 124.1 Anteroposterior (A) and lateral (B) radiographs of the C spine at a neutral position showing normal lordosis.

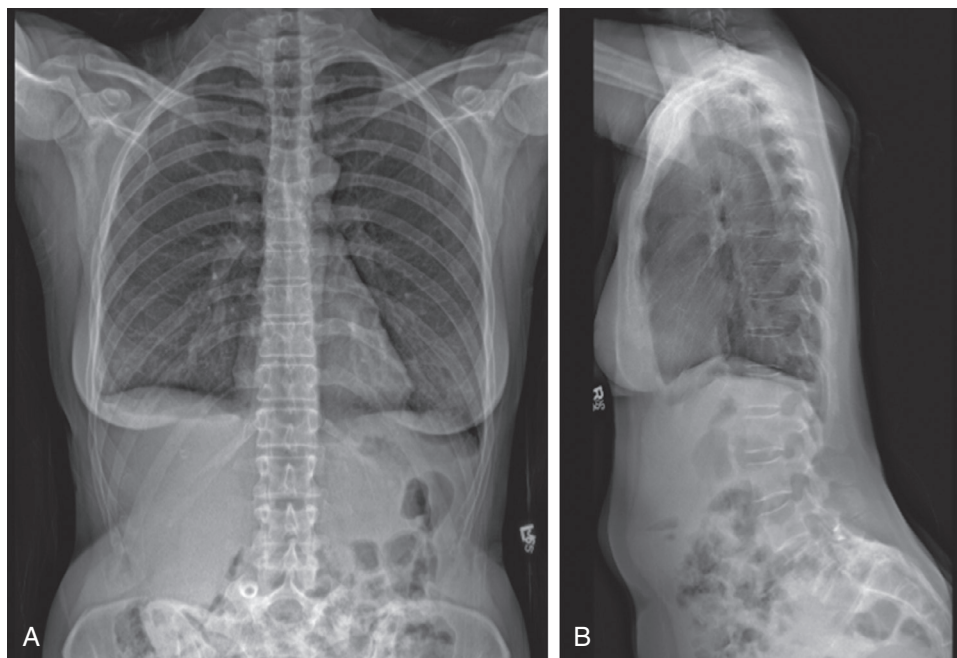


Fig. 124.2 EOS 2D/3D can produce complete anteroposterior (A) and lateral (B) images of the axial skeleton with one shot. Note that compared to the image quality of x-ray images in Fig. 124.1, EOS 2D/3D images are very similar.

lower with EOS than with x-rays. In one study, the mean surface dose with EOS for PA spine was 0.23 milligray (mGy) compared with 1.2 mGy for x-ray. In lateral images, the mean dose was 0.37 mGy compared with 2.3 mGy with x-ray.⁸ Recently, EOS imaging has launched a micro-dose protocol that reduces patient dose by a factor of 5.5.⁵ Even though absolute radiation dose is decreased, the lifetime health benefits of reduced radiation dose from EOS are small. The EOS system is costly and its cost effectiveness is limited by the number of patients it is used for in a given time period. Thus far, EOS has not shown to be cost effective relative to x-ray.^{5,9}

Similar to CR, the EOS system is a tool for the imaging of the skeleton only. EOS images can be acquired within 10 to 30 seconds and can be done while the patient is standing in an upright position. 3D images can be constructed within 15 minutes. EOS software helps with 3D modeling of the spine based on anatomical landmarks defined by the reader and the defined clinical parameters. Luminosity and contrast can be adjusted on the software to help with identification of anatomical landmarks.¹⁰ Even though 3D images can be constructed from EOS, the 3D images are not comparable to the 3D images obtained from computed tomography (CT) as it does not provide information on soft tissues such as muscles, spinal cord, nerves, and viscera. EOS, therefore, is not a good imaging technique for the degenerative spine. In one study, anterior and posterior margins of intervertebral disks were visible in 22% and 64% of cases with EOS, respectively, compared with 84% and 97% with magnetic resonance imaging (MRI).¹¹

Computed Tomography

Like CR, CT uses ionizing radiation, but the radiation doses tend to be very high relative to radiographs. Unlike radiography, this modality acquires images in the axial plane, produces cross-sectional images, and allows for sagittal and coronal 3D reconstructions via post image acquisition processing. Multidetector row CT can acquire all the necessary data for a chest and abdomen study during a single-breath hold, or 15 seconds. CT is the examination of choice for assessing the bony structures of the spine, but the soft tissue assessment is often limited and

requires augmentation with a contrast medium. CT is usually part of trauma and tumor staging protocol because it is relatively fast to acquire images, and it is usually not limited by patient medical conditions. Trauma protocols usually image the whole spine including the skull. CT is the imaging of choice for tumor assessment. In tumor protocol, usually the thoracic, lumbar, and sacral spine are imaged as part of the CT of chest, abdomen, and pelvis.¹²

For CT imaging, the patient is usually in a supine position to ensure limited movement of the spine associated with breathing. CT is sensitive to motion and distorts images due to metal implants. Therefore these two factors can be a limiting factor in obtaining imaging. Since metal implants in the spine are usually made of titanium, routine CT protocols are usually sufficient to obtain good-quality images in the presence of the implants. CT is sensitive for reactive bone change in infection. However, because of the high radiation dose associated, when feasible, MRI should be considered for better assessment of soft tissue abnormalities.¹² The radiation dose for a chest abdomen and pelvis CT is about 20 mSv compared to about 1.5 mSv for a PA spine x-ray. It is unclear what amount of radiation dose increases the risk of cancers, but the range is placed above 200 mSv.¹³

High-contrast resolution and multiplanar reconstruction—the most important advantages of CT—permit an excellent evaluation of the spine for accurate characterization of the osseous details of a lesion, the degree of bony destruction, and spinal alignment (Fig. 124.3). 3D reconstructions can assist in careful fracture evaluation, preoperative planning, and the examination of complex deformity.^{14,15} Although CT is effective for bony degenerative change, it is not the optimal imaging choice for degenerative disk disease. Intravenous contrast can enhance the diagnosis of disc herniation by providing improved delineation of the soft tissue. Finally, CT plays an important role in myelography for demonstrating the outline of nerve roots, and cauda equina (see below). CT-guided interventions, such as biopsies of various lesions, nerve root, facet joint blocks, epidural injection, and even screw placement in spinal surgery, is an integral utility of the technology.

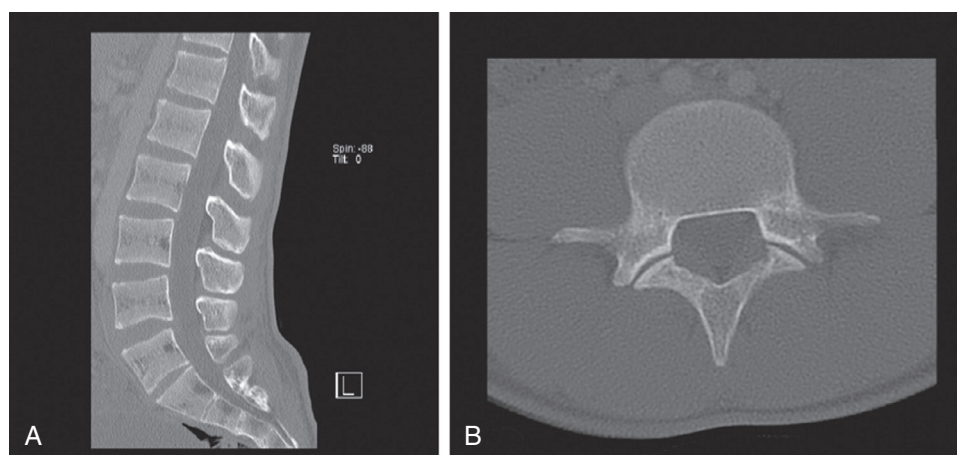


Fig. 124.3 Sagittal reconstructed (A) and axial computed tomography (B) images through L-Spine. The soft tissue is poorly differentiated but bony structures are visualized with great detail.

Magnetic Resonance Imaging

MRI generates multiplanar images that have excellent anatomic and spatial resolution but, unlike CR and CT, it does not involve ionizing radiation. MRI relies on the response of hydrogen nuclei in intracellular and extracellular fluids to a magnetic field generated by the machine. MRI starts by generating an artificial magnetic field to orient the atomic nuclei in one direction. Radiofrequency pulses are then applied to change the direction in which the nuclei are pointing. When the pulse is removed, the nuclei shift back to their steady-state position. Nuclei from fluids of different cells revert to steady state at different rates, ranging from tens of milliseconds to seconds. The translation of the released energy creates discrete regions of varying signal intensities or brightness on the images. The differences in relaxation timing allow the system to distinguish between tissues. The intensity of the signal depends on the number of protons within different tissues, that is, the water content of those tissues. The tissues release the absorbed energy at different rates, which are distinguished as T1 or T2.¹⁶ T1 images reflect the time it takes for nuclei to return to realign with the main magnetic field after the radiofrequency pulse is stopped. T2 reflects the time it takes for nuclei that are spinning to lose their excitation.

An MRI sequence is the combination of radiofrequency pulses, various pre-determined parameters, and magnetic gradients that produce an image with a particular intensity of each tissue type. For the orthopedic surgeon, understanding T1-weighted, T2-weighted, and proton density (PD)-weighted images are the most critical MRI sequences. T1-weighted images, also referred to as T1WI, can be thought of as the most anatomical of images; these images most closely resemble the color gradients of tissue macroscopically. In T1-weighted image (T1WI), fat is bright and the cerebrospinal fluid (CSF) cortex, tendons, and ligaments are dark. It is useful for focusing on anatomical details because the dark CSF, tendons, ligaments, and cortex are surrounded by the bright fat. These are most useful for hemorrhage, marrow replacing masses, or identifying lipoma lesions. Edema and fluid are not well characterized on T1WI.¹⁶ T2-weighted images are best suited for the analysis of pathology related to soft tissue,

bone, ligaments, tendons, joints, bursa, synovium, and muscle. In T2-weighted images, CSF, edema, synovial fluid, fluid cyst, and fresh blood appear bright. On the other hand, old blood, bone, muscle, tendons, white matter, appear dark to gray.¹⁷ Finally, PD-weighted images are the third type of MRI sequence that are essential for an orthopaedic surgeon. PD is an intermediate sequence with some aspects of T1 and others of T2. PD is the ideal sequence for the assessment of joints because it can distinguish between fluid, hyaline cartilage, and fibro cartilage. Fluid (CSF, joint fluid) and fat appear white while muscle, cartilage appear gray to dark. PD was found to be superior to T2-weighted images for detecting lesions in the cervical spinal cord.¹⁸

The fundamental advantage of MRI is its ability to provide a high-resolution depiction of osseous and soft-tissue structures. With respect to the spine, MRI provides excellent visualization of the vertebral body, intervertebral discs, spinal canal, posterior elements, ligaments, paraspinal muscles, nerve roots, and the spinal cord (Fig. 124.4). With multiplanar imaging and the use of various pulse sequences, MRI facilitates abnormality characterization and has been shown to have high sensitivity and specificity for the detection of various disease processes (e.g., 93% and 94%, respectively, for vertebral osteomyelitis¹⁹). MRI has also proved invaluable in the assessment of neoplasms of the spine, with a greater accuracy than CT.²⁰ With regard to the degenerative spine, MRI allows for excellent evaluation of the degree of central and foraminal stenosis as well as the degree of other degenerative changes such as facet arthropathy and degenerative disc disease.

The disadvantages of using MRI include the inability to scan patients with cardiac pacemakers or other embedded ferromagnetic material.²¹ However, new technology is underway to produce implants and pacemakers that are MRI compatible.²² MRI can produce a limited quality image for patients with instrumentation because of metal artifact, even with less ferromagnetic metals (such as titanium) that produce less artifact.²³ Additionally, there are often problems in obtaining a scan in patients with claustrophobia and, compared with CT, MRI has an inferior ability to assess the detail of osseous or calcified structures. Relative contraindications to MRI include the first trimester of pregnancy.

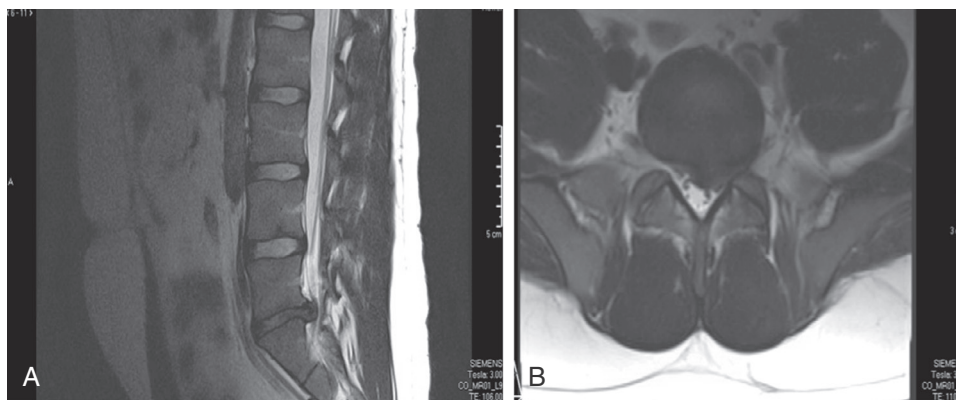


Fig. 124.4 Sagittal (A) and axial (B) T2 magnetic resonance imaging images show the soft tissue structures in the lumbar spine. Notably, disc herniation can be appreciated at the L5-S1 disc. The axial image shows that the disc herniates more towards the right than the left.

Myelography

Myelography is radiography of the spine after injection of a nonionic contrast material into the subarachnoid space, via a lumbar or cervical puncture. The major indications for its use include the imaging of a patient for whom MRI is contraindicated (because of claustrophobia, the presence of pacemaker, etc.), the degradation of image quality in the presence of spinal hardware or cases where kyphoscoliosis makes MRI acquisition challenging and interpretation difficult. Even with the above noted disadvantages of MRI, except for those situations in which metal artifact is a factor, MRI is superior to myelography for the evaluation of spinal abnormality, and has been shown to have a substantially greater accuracy in the detection of herniated disks and a lower false-positive rate.²⁴⁻²⁶ Myelography was the diagnostic test of choice in the 1980s and over time it was superseded by MRI. Between 1999 and 2009, its use declined by almost half, largely owing to the accessibility, availability, and comprehensive imaging obtainable via an MRI.²⁷ Studies have shown the superiority of myelography for the evaluation of some diagnoses such as nerve root compression. In one series of surgically confirmed patients with nerve root compression, MRI underestimated the compression by about 30% compared to only 5% to 7% by myelography.²⁸ Similarly, myelography can be more reliable when deciding the levels for decompressive lumbar surgery, and MRI can underestimate spinal canal and the foramina width.²⁹

In myelography, a water-soluble nonionic contrast agent, such as iohexol (Omnipaque) and iopamidol (Isovue), is injected usually at L2/L3 level under real-time CR, which is called fluoroscopy. Using fluoroscopy allows the identification and correction of an accidental injection into the epidural space and to check whether contrast is obstructed. The fluoroscopy is then used to obtain images. Alternatively, in most cases, CT images are obtained. These images are evaluated for evidence of compression of the CSF column, disc herniation or compression, bony osteophytes, spinal stenosis, intramedullary tumors, nerve root compression, or meningitis.^{27,30} Myelography, in conjunction with CT, affords an excellent evaluation of foraminal stenosis and stenosis adjacent to spinal instrumentation such as pedicle screws.

In the degenerative lumbar and cervical spine, it is important to carefully trace the pathway of each nerve root sleeve to evaluate for foraminal stenosis and to carefully examine the axial CT images to evaluate for central stenosis. Compared with MRI, the use of myelography is not as common. For this reason, less experienced clinicians may not be familiar with the evaluation of these imaging studies. Nevertheless, all clinicians should attempt to evaluate the postmyelography imaging studies and then correlate their impressions with the radiology reports. In patients for whom spinal surgery is being considered, a clinician may also consult with a radiologist experienced in this technique.

Disadvantages of myelography include the potential for an allergic reaction to the contrast agent, the use of ionizing radiation, a lower seizure threshold, bleeding, risk of infection, headache, nausea, the invasiveness of and pain associated with the

examination, the time and expertise needed to perform the study, the risk of neural damage, and the inability to determine areas of compression below the blocks in contrast flow.³¹⁻³⁴ The primary contraindication to myelography is allergy to the nonionic contrast agent. Relative contraindications include seizure disorder, bleeding disorders, concurrent use of anticoagulants, infection or skin disease over the site of injection, and pregnancy.

Nuclear Scintigraphy

Nuclear scintigraphy is a quick, relatively inexpensive, widely available, and sensitive test used for the evaluation of musculoskeletal pathology. Unlike the previously mentioned imaging modalities, nuclear scintigraphy (also termed radionuclide bone scan) provides anatomic and physiologic information via the administration of a radiopharmaceutical compound, usually technetium-99m-labeled diphosphonates, into a patient's venous system. The radioisotope is preferentially deposited in regions of increased bone remodeling or activity. It usually takes 2 to 6 hours after injection for 50% of the dose to be deposited in the skeletal system.³⁵ The deposition of the radioisotope allows areas of increased or decreased bone turnover to be differentiated by a scintillation camera that detects and localizes the gamma radiation emitted by the injected agent. The radiotracer uptake depends on blood flow and the rate of new bone formation. Depending on the purpose of the bone scan, different imaging protocols are used.

The commonly used "three-phase" radionuclide bone scan acquires images at different postinjection times. In phase 1 (blood-flow phase), 2- to 5-second images are obtained during the first minute after injection. This demonstrates the perfusion characteristics and the quality of blood flow to a certain area. For phase 2 (blood-pool or soft-tissue phase), images are obtained 5 to 10 minutes after injection aimed at understanding the pooling of blood since inflammation can increase blood collection in the infected area. In phase 3 (delayed or static bone phase), the radiotracer usually deposits into bone mineral, and images are generally taken 2 to 3 hours post-injection. The degree of uptake depends on the blood flow and the rate of new bone formation. In some cases, a phase 4 static image is taken after 24 hours to assess for overall distribution of the radiotracer (Fig. 124.5). In osteomyelitis, for example, there is focal hyperperfusion, focal hyperemia, and focally increased bone uptake. The sensitivity for detection of osteomyelitis by nuclear scintigraphy ranges from 70% to 100% when increased uptake is noted on all three phases.³⁶⁻³⁹

This modality is particularly useful for the evaluation of metabolic bone disorders, stress fractures, primary and metastatic neoplasms, infections, and degenerative disorders.³⁹⁻⁴³ Other major advantages of nuclear scintigraphy are that it can evaluate the complete skeleton in one study, and can detect osseous lesions early because of its high sensitivity. These characteristics make scintigraphy ideal for surveying the skeleton for metastatic disease, which is one of the most common indications for this modality. Bone scans are extremely sensitive—more sensitive than CR—for detecting skeletal abnormalities due to metastatic disease. In 75% of patients with malignancy and bone pain, a bone scan is usually abnormal.³⁵ Nuclear scintigraphy also can be used in



Fig. 124.5 DEXA scan in a 52-year-old male shows the increased signal intensity in regions with higher uptake of the radioisotope, notably the pelvic rim, physeal plate of the distal femur, proximal radius, and shoulder region. The radioisotope is cleared by the kidneys and hence the increased uptake in the bladder.

lieu of MRI to help determine the age of vertebral compression fractures.

Nuclear scintigraphy has several disadvantages. Most importantly, it is relatively nonspecific; any condition that causes increased bone turnover will appear similar. Furthermore, aggressive lesions that do not allow time for remodeling, or osteoblastic lesions with a low metabolic rate, can generate a falsely negative study.⁴¹ It also requires a substantial length of time to complete the study and has poor spatial resolution. However, single-photon emission computed tomography (SPECT) can improve the spatial resolution of nuclear scintigraphy because it helps to detect osseous lesions in more complex anatomical regions, such as the spine and pelvis. SPECT (see below) has improved the ability to detect vertebral metastases and to localize lesions within different areas of the vertebra, including the pars intra-articularis in patients with suspected or known isthmic spondylolisthesis.^{44,45}

Functional Metabolic Imaging

Functional metabolic imaging is an imaging technique that aims to assess the physiologic activity in the bone or soft tissue by measuring changes in blood flow, chemical composition, uptake of glucose, or changes in metabolism. Unlike the imaging techniques discussed earlier, functional imaging focuses on exploring the physiologic activity of a tissue rather than elucidating the anatomical details of a structure. Positron emission tomography (PET) is another form of nuclear medicine imaging modality and is a type of functional metabolic imaging that is used in the evaluation of physiologic changes in various tissues. In its most common form, 18-F-labeled 2-fluoro-2-deoxyglucose (FDG) is used as the positron-emitting radionuclide tracer that emits gamma radiation at levels proportional to intracellular glucose metabolism.⁴⁶ The concentration of FDG is proportional to the level of metabolic activity of the tissue. FDG usually takes about 30 minutes to 1 hour for systemic incorporation into the various body tissues.⁴⁷ Therefore a scan is done usually 30 minutes after injection. Increasingly, PET is combined with a CT in the same machine, and the patient gets a PET and CT in one visit. CT allows the reconstruction of a 3D image series that provides anatomical detail, whereas PET collects data on metabolic information. As a result, one image series is used for anatomical and metabolic assessment of the patients.⁴⁸⁻⁵⁰

The primary indications for this modality are the detection of primary and metastatic neoplasms and the evaluation of tumor response to treatment. Because neoplasms often have higher rates of metabolic activity than normal surrounding tissue, a PET scan can identify tissue with higher metabolic activity. The applications of PET scanning for musculoskeletal conditions continue to grow: differentiation between osteoporotic and pathologic vertebral fractures, between aseptic loosening and infection in knee and hip arthroplasty, and between benign and metastatic spinal metastases.^{51,52} The principal disadvantages of a PET scan are its limited availability and high cost.

Single Photon Emission Computed Tomography

SPECT/CT is another form of functional metabolic imaging that relies on a radionuclide for imaging of the spine. SPECT uses a gamma ray camera detector to capture gamma rays released by an injected radioisotope material. PET and SPECT have some differences that make SPECT a more suitable imaging technique for many different imaging needs.⁵³ SPECT is cheaper relative to PET partially because it can use longer lived and easily obtainable radioisotopes. SPECT uses technetium-99m, iodine-123, or iodine-131 radioisotopes as opposed to the fluorine-18 isotope used in PET. Relative to PET, SPECT has a lower contrast and spatial resolution (maximum resolution of about 1 cm). Similar to a PET, SPECT can be combined with a CT, (SPECT/CT) to provide functional information from the SPECT and anatomical information from the CT. Combining the CT is helpful because it can also improve the attenuation and the quality of the photon emission data.^{54,55} As a result, a single imaging technique can be used to acquire high-quality functional and anatomical information.

SPECT/CT has numerous uses in the evaluation of orthopaedic pathologies, such as fractures in the spine, compression fractures of the vertebra, evaluation of malignancies, infections, bone trauma, and identifying degenerative changes. For example, using SPECT/CT fractures of pars interarticularis can be distinguished from areas of increased activity such as facet joint arthritis.⁵⁶ There is strong evidence for the higher sensitivity of SPECT/CT compared to SPECT alone or nuclear scintigraphy for the detection of bone tumor. In one study, SPECT/CT had a sensitivity of 100% for the detection of cancerous lesions in the spine compared with 64% for nuclear scintigraphy and 86% for SPECT alone.^{57,58}

Interventional Radiology Techniques

Interventional radiology (IR) is a subspecialty of radiology in which fluoroscopy, CT, or MRI are used to guide a wide range of percutaneous treatments throughout the body, including the musculoskeletal system. Under image guidance, tissue samples can be obtained that can lead to diagnosis and, in turn, can guide or directly assist with therapeutic intervention, for example, image-guided biopsies and aspirations, radiofrequency ablation (RFA), vertebroplasty, kyphoplasty, and angiography.

IR has played an increasingly dominant role in the management of previously surgically treated diseases. Image-guided RFA is used to treat benign and metastatic bone lesions. Advantages of RFA over surgery include short procedures, the possibility of outpatient treatment, lower morbidity, and decreased cost. In vertebroplasty and kyphoplasty, fluoroscopy, or CT guidance can facilitate an accurate transpedicular approach and guide cement placement for treatment of painful lesions of the spine.⁵⁹ Angiography, which is often used for the evaluation of vascular tumors, can be combined with embolization before surgical intervention. Angiography and embolization are especially useful for hypervascular tumors, such as aneurysmal bone cyst, giant cell tumor, and angiosarcoma. Preoperative arterial embolization improves visibility during surgery, and permits safer and faster surgery.⁶⁰ In some cases, IR is a safer and more effective procedure than surgical intervention. Osteoid osteoma was historically removed with wide resection to excise the small nidus. Surgery was found to be associated with a higher risk of fracture and longer immobilization. In the last 10 years, percutaneous CT-guided ablation has become increasingly popular because of the high success rate and the low morbidity. CT-guided ablation can be used in many parts of the skeleton, except when the osteoma is close to neurovascular structures.⁶¹

IMAGING OF THE DEGENERATIVE SPINE

The primary reasons for presentation to a spine care specialist are conditions such as low back pain, neck pain, numbness and tingling in legs, and limited mobility in the spine. Degenerative diseases of the spine are one of the most common presenting diagnoses for patients. Such patients often present with neck pain (alone or in conjunction with radicular arm pain), back pain (alone or in conjunction with radicular leg pain), signs and symptoms of myelopathy, or signs and symptoms of neurogenic claudication.

For patients with benign neck or back pain, 70% to 90% show improvement in symptoms within 1 month after initial presentation.^{62,63} Therefore imaging in the form of conventional radiographs is usually delayed for 6 weeks after initial presentation in the absence of “red flags” (e.g., fever, chills, trauma, night or rest pain, weight loss, history of malignancy, or neurologic deficit). However, because patients frequently present initially to their primary care physician or to the emergency department, 6 weeks often can elapse by the time they are evaluated by a spine specialist. For this reason, many specialists obtain conventional radiographs on an initial evaluation.

For a patient with degenerative changes of the cervical or lumbar spine, initial radiographs should be scrutinized for the presence of degenerative changes. Evidence of spondylosis is seen in most spines of elderly patients.⁶⁴⁻⁶⁶ After conventional radiographs have been evaluated, the patient is often referred for nonoperative treatment, including antiinflammatory medications, physical therapy, and interventional pain management as needed. MRI studies are not usually indicated as one of the initial studies, but may be considered if the symptoms persist. As described above, MRI is the imaging modality of choice for the degenerative spine because it provides excellent multiplanar (axial, sagittal, and coronal) evaluation and facilitates careful evaluation of spinal alignment and morphology of the vertebral bodies and disc. Most importantly, with regard to the degenerative spine, this imaging modality helps evaluate for the presence or absence of spinal stenosis.

An MRI imaging study of the spine is best evaluated by using a systematic approach. One such approach is described here⁶⁷:

1. Determine which pulse sequences and specialized MRI imaging studies are available for review. This includes identifying whether it is a T1-weighted image, a conventional T2-weighted image, or PD. Usually, MRI of the spine includes images in the sagittal and axial planes.
2. Evaluate T2-weighted (often sagittal plane) images for recognition of areas of increased T2-weighted signal that are not physiologic or expected. Disc herniation can be identified by following the bright high-intensity signal of the CSF.
3. Evaluate T1-weighted images for improved detection of anatomic detail and the correlation of the disturbance in local and regional anatomy on the T1-weighted images with areas of increased signal intensity on the T2-weighted images.
4. Evaluate specialized MRI pulse sequences that may be specific to the region or the disease process of interest such as PD or fat-suppressed T2-weighted images. The pulse sequences will be selected based on the pathology being evaluated.
5. Correlate the above imaging information with the patient's history, physical examination, and laboratory study results to determine the most likely differential diagnostic considerations.

For the evaluation of degenerative disc disease, the degree of stenosis should be evaluated on the MRI study. Although there are several objective measures of stenosis, most clinicians tend to use the words “mild,” “moderate,” and “severe” based on their own criteria. More objective measures of stenosis include a definition for the lumbar spine as a cross-sectional area less than

100 mm².⁶⁸ On CR, relative stenosis in the cervical spine is defined as an AP canal diameter of less than 13 mm, and absolute stenosis is defined as an AP canal diameter of less than 10 mm.⁶⁹ Although less experienced clinicians tend to evaluate stenosis only on the sagittal images, we suggest the additional use of axial T2-weighted images to help evaluate for stenosis in the central canal, the lateral recess, and the neural foramina.

Degenerative disc changes of the spine are common with up to 70% to 80% of the population showing evidence of disc degeneration by the time they are 50. The elderly have especially high rates of degenerative changes of the spine, although many patients with such degenerative changes may be asymptomatic.⁷⁰⁻⁷² For example, Boden et al. found that 28% of 67 asymptomatic patients had abnormal MRI findings suggestive of a herniated nucleus pulposus or lumbar stenosis. Additionally, those who were more than 60 years old had a higher incidence of degenerative findings. Of the 60- to 80-year-old age group, 93% had MRI evidence of degenerated discs: 79% with bulging discs and 36% with disc herniations.⁷⁰ In a similar study of the cervical spine, Boden et al. found that 28% of 63 asymptomatic patients older than 40 years had abnormal MRI findings (disc herniations, bulging discs, and foraminal stenosis). When evaluating for disc space narrowing or signs of degeneration, nearly 60% of 63 asymptomatic patients older than 40 years had positive findings.⁷¹

IMAGING STUDY INTERPRETATION

As with all diagnostic studies, there are several limitations with regard to imaging the spine that should be considered. These limitations can be divided into the following broad categories: (1) lack of correlation of the imaging findings with the patient's history and physical examination; (2) lack of correlation of imaging findings with findings on other imaging modalities; (3) poor understanding of normal anatomic variance; and (4) abnormal anatomic and physiologic findings that are common in asymptomatic patients.

Other potential limitations relate to the practice of clinicians providing the formal readings for imaging studies. It is important to note that there is a tendency for specialists to focus on their area of interest when evaluating various imaging studies. For example, an orthopaedic surgeon evaluating AP and lateral radiographs of the lumbar spine may focus on the spine, which could result in missing other findings such as bowel obstruction, nephrolithiasis, gallstones, or a lytic lesion of the ileum. If the clinician is solely responsible for evaluating the imaging studies, then he or she must completely evaluate all images and all aspects of the images for musculoskeletal and nonmusculoskeletal-related diagnoses. In such cases, the clinician must carefully and critically evaluate all imaging studies using a systematic approach. For example, instead of selecting a few radiographs from the jacket or a few images on the workstation to make the diagnosis of L4-L5 grade-1 spondylolisthesis with moderate stenosis, the clinician must evaluate all pulse sequences and every single image to rule out additional findings. Reading such films in detail requires additional time on the part of the clinician, but the team approach that combines the expertise of a trained

radiologist with clinical patient information provided by the clinician can help ensure an accurate diagnosis and can guide treatment.

CONCLUSION

Conventional and advanced imaging studies, along with a comprehensive history and physical examination, are essential in the diagnosis and treatment of patients with known or suspected spine pathology. There are a wide variety of available imaging techniques with which the surgeon should be familiar, which include conventional radiographs, EOS 2D/3D, CT, MRI, myelography, nuclear scintigraphy, PET/CT, and SPECT/CT. As these techniques evolve with time, and as new imaging modalities are introduced, it is important for surgeons to stay current with the changing capabilities and limitations of each technique. Because of the increasing sophistication and complexity of the imaging modalities, it is important for the clinician to actively collaborate with their radiologist colleagues to understand all the details and nuances of a particular technique. The interpretation of imaging studies in consultation with colleagues in radiology, when needed, is crucial. Such consultation, and a team approach, helps ensure that accurate and complete information is obtained from the imaging studies, which can be correlated with each patient's individual clinical profile.

FUTURE CONSIDERATIONS

One of the primary areas for advancement in spine surgery is the use of intraoperative imaging, especially in conjunction with minimally invasive procedures, or for patients with advanced deformity.

Currently, several types of imaging studies can be obtained in the operating room. The existing modality of choice is fluoroscopy, which provides real-time image acquisition. However, it involves a substantial amount of radiation and does not provide imaging in the axial plane. Several systems now offer the acquisition of CT images, which provide axial imaging, reconstructed sagittal and coronal imaging, and 3D volumetric imaging. Intraoperative MRI and MRI-compatible operating room suites are also available in selected centers in the United States. Although the use of advanced imaging modalities in the operating room may improve outcomes and patient safety, there remains room for the optimization of current-generation systems in terms of ease and efficiency of incorporation into operating room workflow and overall quality, resolution, and accuracy of the images.

As with any technology, imaging systems will continue to evolve and improve. Other potential developments that will affect future imaging of the spine include the increased speed of CT and MRI with newer generations of these systems, multiple-access imaging, decreased cost, and improved image resolution (especially MRI, with the introduction of 3.0-T systems). The clinician will continue to be responsible for keeping abreast of these new technologies, critically evaluating the clinical efficacy and cost effectiveness of imaging, and safely incorporating the modalities into clinical practice.

For a complete list of references, go to [ExpertConsult.com](https://www.expertconsult.com).

SELECTED READINGS

Citation:

Hartley KG, Damon BM, Patterson GT, et al. MRI techniques: a review and update for the orthopaedic surgeon. *J Am Acad Orthop Surg*. 2012;20(12):775–787.

Level of Evidence:

IIIa

Summary:

MRI is one of the most important imaging techniques at the disposal of orthopaedic surgeons. This article provides an excellent summary of the various sequences used in MRI and the application of these sequences. The authors discuss the appropriate skills needed for the evaluation of pathologies with MRI and the analysis of MRI images.

Citation:

Melhem E, Assi A, El Rachkidi R, et al. EOS((R)) biplanar x-ray imaging: concept, developments, benefits, and limitations. *J Child Orthop*. 2016;10(1):1–14.

Level of Evidence:

IIIa

Summary:

EOS is new tool that is competing with conventional radiograph for the evaluation of spine disease. In this article, Melhem et al. discuss the potential applications of EOS in various musculoskeletal disorders and compare them against conventional radiographs on different parameters (radiation dose, cost, accessibility, image quality).

Citation:

Saha S, Burke C, Desai A, et al. SPECT-CT: applications in musculoskeletal radiology. *Br J Radiol*. 2013;86(1031):20120519.

Level of Evidence:

IIa

Summary:

SPECT/CT is a technology with significant potential. The article by Saha et al. reviews the technique, basic principles, and applications of SPECT/CT. The authors provide guidance on the analysis of SPECT/CT from different regions of the skeleton and provide case series and examples to illustrate their points.

Citation:

Bastiaannet E, Groen H, Jager PL, et al. The value of FDG-PET in the detection, grading and response to therapy of soft tissue and bone sarcomas: a systematic review and meta-analysis. *Cancer Treat Rev*. 2004;30(1):83–101.

Level of Evidence:

IIa

Summary:

PET/CT uptake is commonly used for the evaluation of metabolic activity of malignancies. This systematic review evaluated the evidence for the uptake of FDG in soft tissue and bone sarcomas. The study is significant in highlighting the parameters needed to evaluate when assessing PET/CT studies.

Citation:

Love C, Din AS, Tomas MB, et al. Radionuclide bone imaging: an illustrative review. *Radiographics*. 2003;23(2):341–358.

Level of Evidence:

IIc

Summary:

Bone scintigraphy continues to play an important role in the evaluation of bone density and various pathologic conditions. Love et al. provide a comprehensive review on the evaluation of bone scans and focus on pathologies related to multiple joints. In this article, they include an excellent collection of normal and pathologic images obtained through bone scintigraphy.

Citation:

Ozdoba C, Gralla J, Rieke A, et al. Myelography in the age of MRI: why we do it, and how we do it. *Radiol Res Pract*. 2011;329017.

Level of Evidence:

V

Summary:

The authors provide a comprehensive overview of the current uses of myelography. The focus is on the details of obtaining optimal myographic studies and they discuss the current applications in the era of more advanced imaging techniques. This article provides appropriate guidelines on the analysis of myelographic images.

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